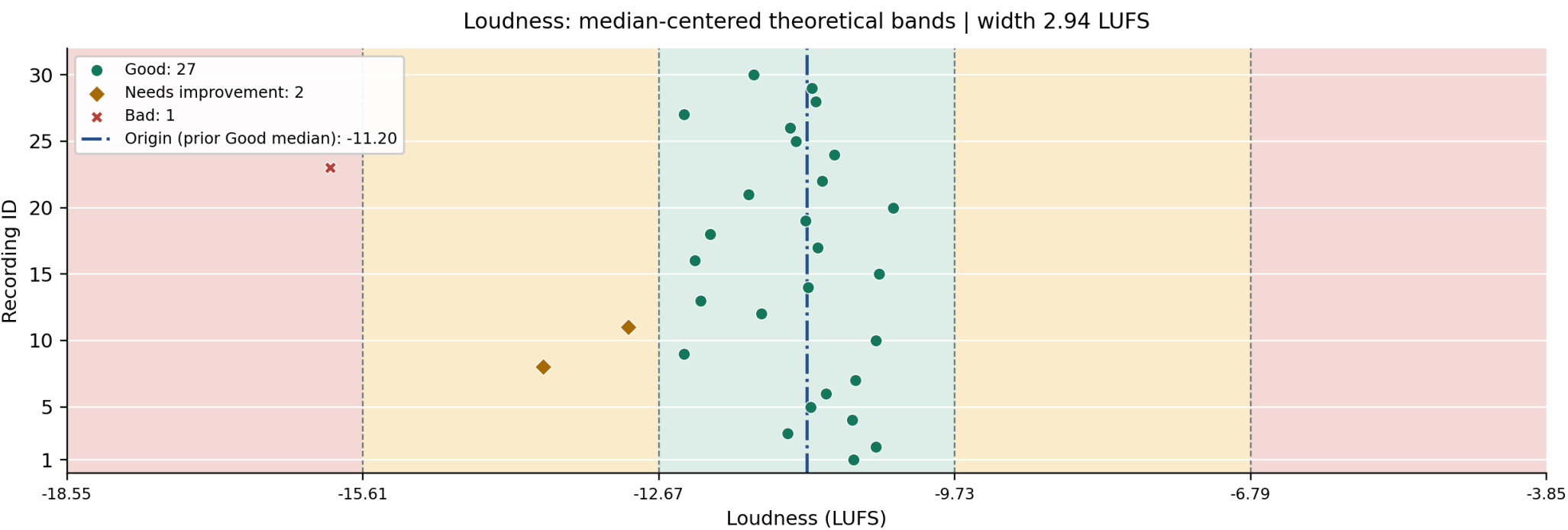


Loudness: median-centered theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / Units: LUFS

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-18.55 to < -15.61	-15.61 to < -12.67	-12.67 to -9.73	> -9.73 to -6.79	> -6.79 to -3.85

Shift all boundaries by median minus old midpoint. Preserve each band width. Displayed values are rounded.



Under these hypothetical boundaries: 27 Good / 2 Needs improvement / 1 Bad

Fixed origin: -11.20 LUFS (median of the prior 26 Good recordings). Good now spans origin +/- 1.47.

Loudness uses LUFS; negative values are valid. Equal-width extensions outside the valid domain are not collection targets.

Good includes both endpoints; the adjacent outer cutoffs belong to Needs improvement. Further tails are Bad in this scenario.

All 30 recordings are reclassified after the shift; 0 are off scale (arrows). The origin is not recalculated after shifting.

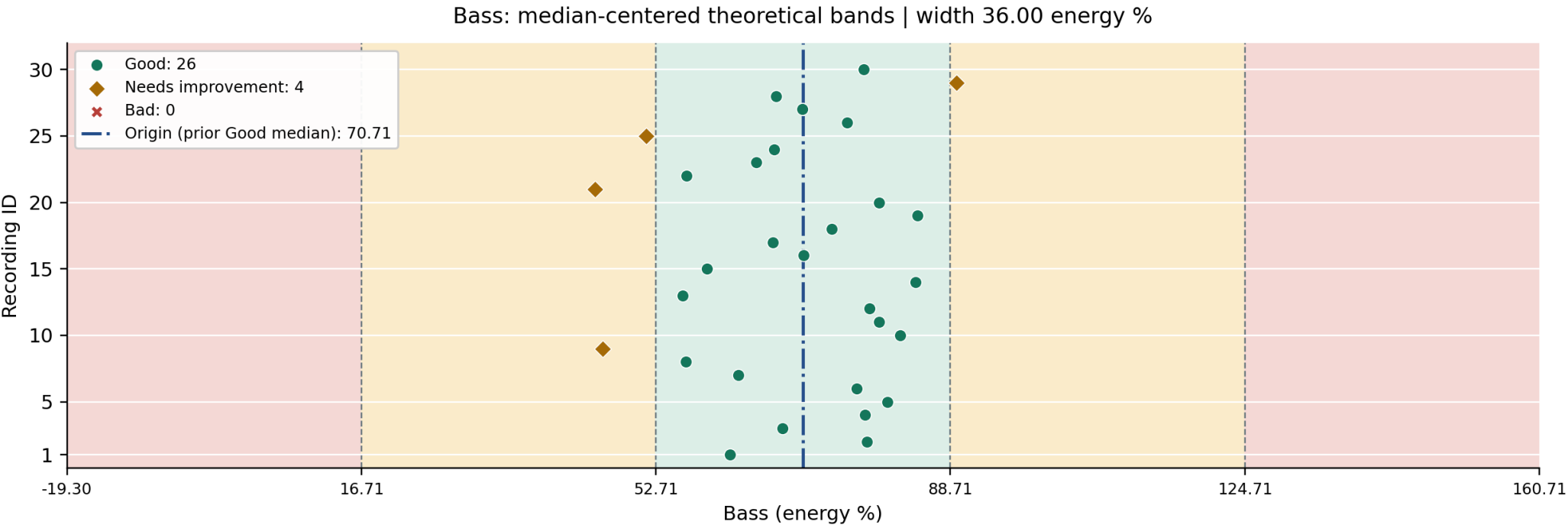
All source recordings were labeled good. These hypothetical classes need independent listener validation; production thresholds are unchanged.

Bass: median-centered theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / Units: energy %

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-19.30 to < 16.71	16.71 to < 52.71	52.71 to 88.71	> 88.71 to 124.71	> 124.71 to 160.71

Shift all boundaries by median minus old midpoint. Preserve each band width. Displayed values are rounded.



Under these hypothetical boundaries: 26 Good / 4 Needs improvement / 0 Bad

Fixed origin: 70.71 energy % (median of the prior 26 Good recordings). Good now spans origin +/- 18.00.

Valid measurement domain: 0 to 100 energy %. Equal-width extensions outside the valid domain are not collection targets.

Good includes both endpoints; the adjacent outer cutoffs belong to Needs improvement. Further tails are Bad in this scenario.

All 30 recordings are reclassified after the shift; 0 are off scale (arrows). The origin is not recalculated after shifting.

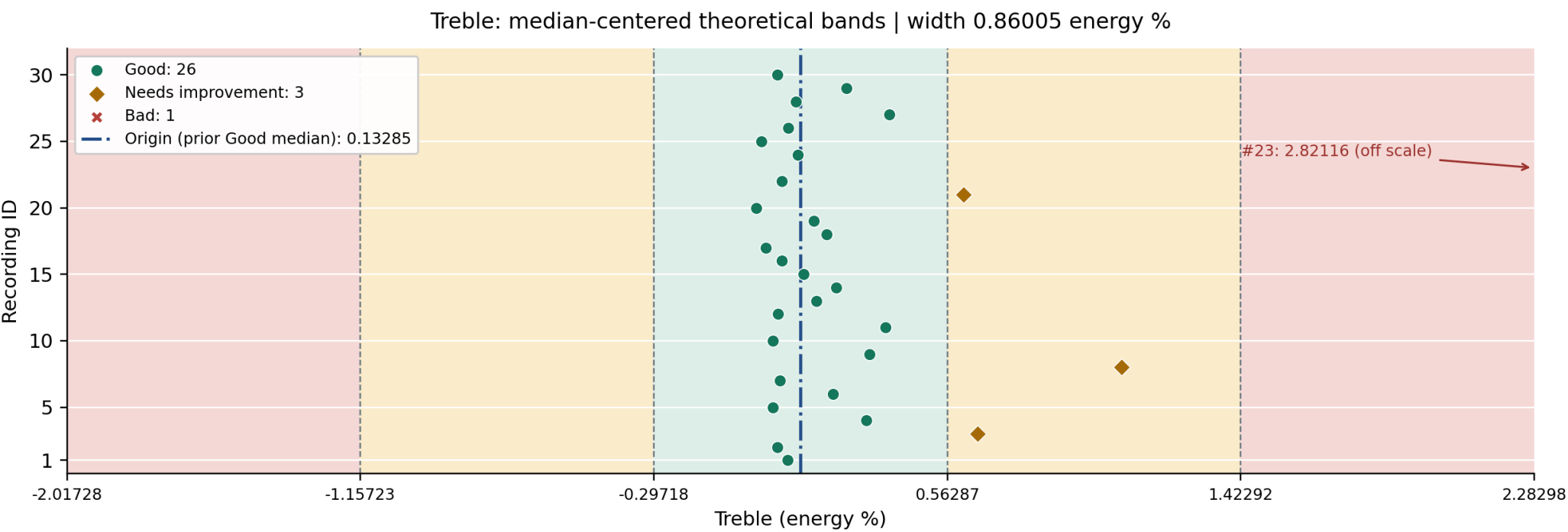
All source recordings were labeled good. These hypothetical classes need independent listener validation; production thresholds are unchanged.

Treble: median-centered theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / Units: energy %

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-2.01728 to < -1.15723	-1.15723 to < -0.29718	-0.29718 to 0.56287	> 0.56287 to 1.42292	> 1.42292 to 2.28298

Shift all boundaries by median minus old midpoint. Preserve each band width. Displayed values are rounded.



Under these hypothetical boundaries: 26 Good / 3 Needs improvement / 1 Bad

Fixed origin: 0.13285 energy % (median of the prior 26 Good recordings). Good now spans origin +/- 0.43003.

Valid measurement domain: 0 to 100 energy %. Equal-width extensions outside the valid domain are not collection targets.

Good includes both endpoints; the adjacent outer cutoffs belong to Needs improvement. Further tails are Bad in this scenario.

All 30 recordings are reclassified after the shift; 1 are off scale (arrows). The origin is not recalculated after shifting.

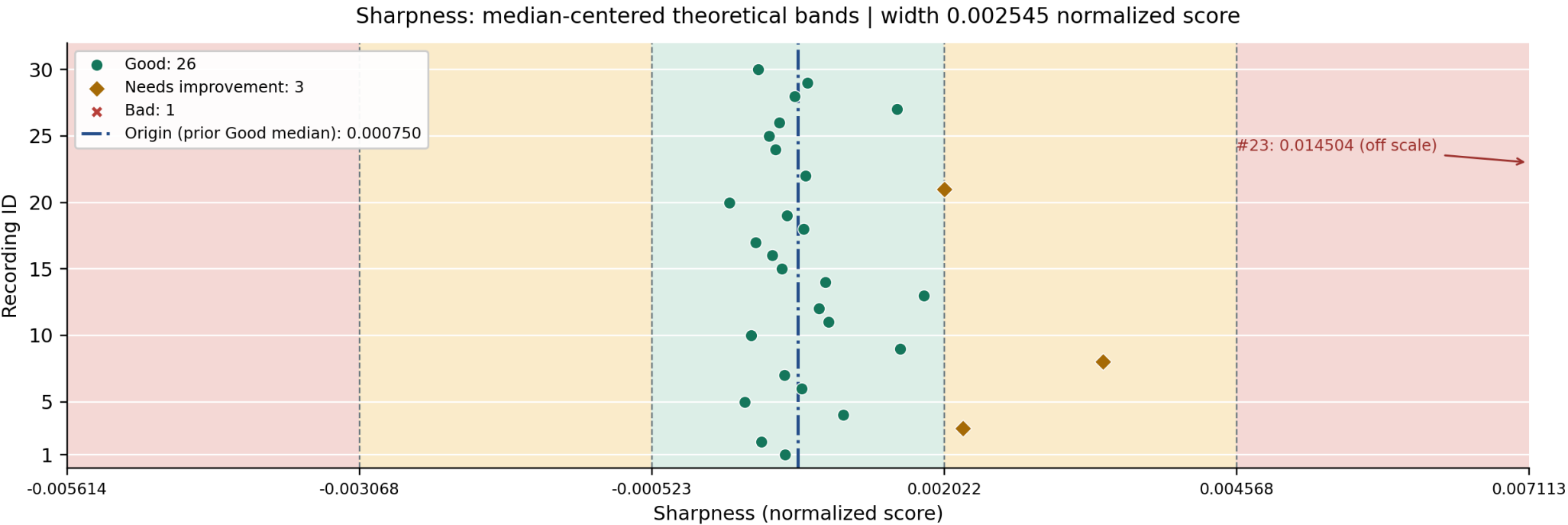
All source recordings were labeled good. These hypothetical classes need independent listener validation; production thresholds are unchanged.

Sharpness: median-centered theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / Units: normalized score

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-0.005614 to < -0.003068	-0.003068 to < -0.000523	-0.000523 to 0.002022	> 0.002022 to 0.004568	> 0.004568 to 0.007113

Shift all boundaries by median minus old midpoint. Preserve each band width. Displayed values are rounded.



Under these hypothetical boundaries: 26 Good / 3 Needs improvement / 1 Bad

Fixed origin: 0.000750 normalized score (median of the prior 26 Good recordings). Good now spans origin +/- 0.001273.

Negative sharpness scores are outside this measurement domain. Equal-width extensions outside the valid domain are not collection targets.

Good includes both endpoints; the adjacent outer cutoffs belong to Needs improvement. Further tails are Bad in this scenario.

All 30 recordings are reclassified after the shift; 1 are off scale (arrows). The origin is not recalculated after shifting.

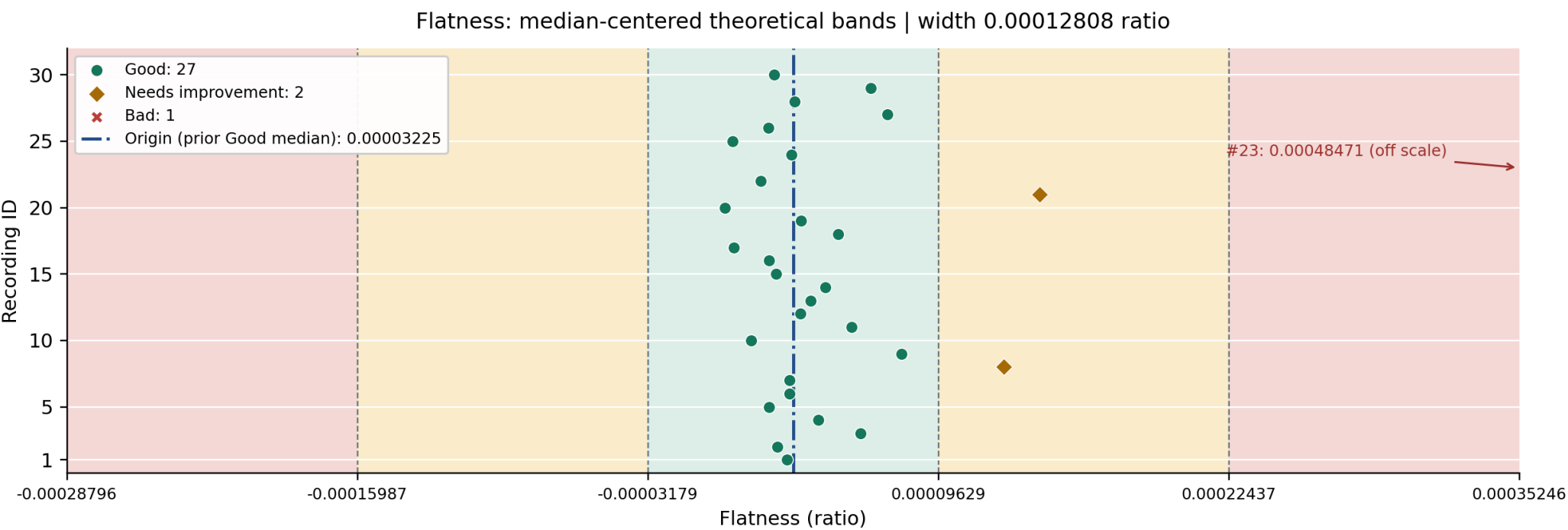
All source recordings were labeled good. These hypothetical classes need independent listener validation; production thresholds are unchanged.

Flatness: median-centered theoretical ranges

KaraOK / Hypothetical sampling scenario / 30 existing recordings / Units: ratio

Leftmost - Bad	Left - Needs improvement	Good	Right - Needs improvement	Rightmost - Bad
-0.00028796 to < -0.00015987	-0.00015987 to < -0.00003179	-0.00003179 to 0.00009629	> 0.00009629 to 0.00022437	> 0.00022437 to 0.00035246

Shift all boundaries by median minus old midpoint. Preserve each band width. Displayed values are rounded.



Under these hypothetical boundaries: 27 Good / 2 Needs improvement / 1 Bad

Fixed origin: 0.00003225 ratio (median of the prior 26 Good recordings). Good now spans origin +/- 0.00006404.

Valid measurement domain: 0 to 1 ratio. Equal-width extensions outside the valid domain are not collection targets.

Good includes both endpoints; the adjacent outer cutoffs belong to Needs improvement. Further tails are Bad in this scenario.

All 30 recordings are reclassified after the shift; 1 are off scale (arrows). The origin is not recalculated after shifting.

All source recordings were labeled good. These hypothetical classes need independent listener validation; production thresholds are unchanged.